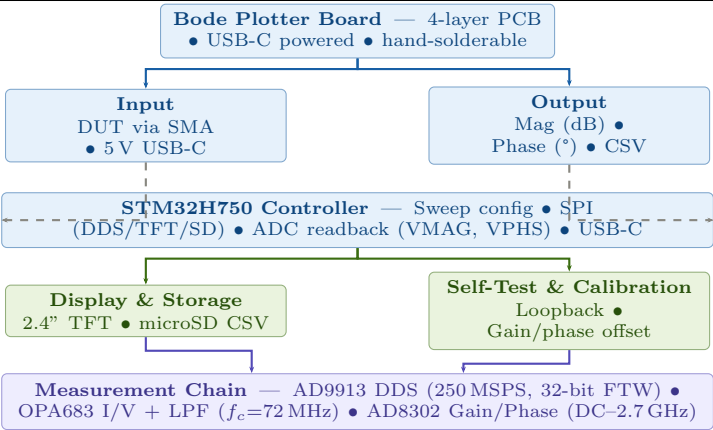


High-Frequency Bode Plotter

10 mHz–100 MHz • ±30 dB Magnitude • ±1° Phase • Standalone • No PC • Open Firmware

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1. Deliverables



2. System Specifications

Frequency Range

10 mHz – 100 MHz • 0.058 Hz resolution (32-bit FTW) • Up to 4096 log-spaced sweep points

Magnitude & Phase

Magnitude: ±30 dB range, ±0.5 dB accuracy (30 mV/dB)
Phase: 0°–180°, ±1° accuracy (10 mV/°)

Signal Source & ADC

AD9913 DDS, 250 MSPS, 25 MHz TCXO reference
STM32H750 480 MHz M7 • 12-bit ADC • 64× oversample • 0.027 dB/LSB

Power

USB-C +5 V, ~300 mA • 3.3 V digital (TLV75533) • 1.8 V analog (TLV75518) • Ferrite-bead isolated rails

4. Interfaces

SPI Bus

AD9913 config • 2.4" TFT (ILI9341) • microSD • touch controller

Analog RF & ADC

IOUT/IOUTB diff. out, 50 Ω DUT port, REF tap to AD8302
VMAG: 30 mV/dB • VPHS: 10 mV/° • RC filter f_c =159 Hz

USB-C

Full-speed USB • Power + data • Future: live streaming

5. Performance

Key Features

- Fully standalone — no PC, no software
- 4096-point log sweep, touch readout of freq/gain/phase
- Real-time Bode plot on 2.4" TFT display
- microSD CSV export, open STM32 HAL firmware

3. Key ICs

AD9913 — DDS Synthesiser

250 MSPS • 10 mHz–100 MHz output • 32-bit FTW (0.058 Hz res.) • Built-in linear sweep • SPI interface

AD8302 — Gain/Phase Detector

DC to 2.7 GHz • ±30 dB magnitude range • 0°–180° phase range • 30 mV/dB • 10 mV/° • Single 5 V supply

OPA683 — High-Speed Op-Amp

I/V conversion (TIA config) • f_c = 72 MHz LPF • 50 Ω output conditioning

STM32H750VBT6 — MCU

480 MHz Cortex-M7 • SPI ×4 (DDS/TFT/SD/touch) • USB-C Full Speed • LQFP-100, hand-solderable

TLV75518 / TLV75533 — LDOs

1.8 V analog rail + 3.3 V digital rail • 1 A rated • Separate pour planes.

6. Cost Breakdown

Item	INR
PCB Fabrication	2,400
PCB Assembly	2,680
Components / BoM	9,949
Total	15,029

Further details: [Bode Plotter](#)